Preserving Exact Long-Range Recall in Compressed Transformers

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October 2026

Abstract

Serving large language models across long context horizons is critically bottlenecked by the Key-Value (KV) cache, whose linear footprint in sequence length T exhausts accelerator high-bandwidth memory (HBM) and restricts multi-tenant throughput. Post-training compression strategies have split into two competing paradigms: compiling historical context into fixed-size recurrent matrix states, or selectively evicting tokens from the KV cache. In this work, we demonstrate the fundamental geometric failure modes of both approaches and introduce a principled mechanism that resolves them without fine-tuning. First, we prove the **Rank-r Spectral Dilution Horizon**: any fixed-size recurrent state acts as a low-pass spectral filter whose energy allocation to an individual token decays as $\mathcal{O}(1/T)$, causing exact long-range recall to collapse to 0.0% beyond the local window despite capturing over 95% of attention output variance ($R^2 = 0.9505$) and preserving 100% general reasoning accuracy. Second, we uncover the **Query-Answer Asymmetry Principle**: existing selective token eviction schemes fail on complex entities because causal prefix scoring is blind to terminal prompt queries, while greedy top-k selection retains query-matching cue words while severing the adjacent multi-token payload. In response, we introduce **HALO** (**H**ybrid **A**ttention with **L**andmark **O**bservation-diffusion), which pairs an exact local sliding window with phase-rectified observation alignment, bottom-up geometric surprise, and an **Exponential Salience Halo** ($\kappa(\delta) = \exp(-0.10|\delta|), K = 16$) that diffuses importance contiguously across complete factual clauses, alongside a unified **HALO + BLUR** synthesis combining exact landmarks with compressed residual background statistics. Evaluated across three distinct foundation model families from independent organizations—Alibaba (`Qwen3.5-0.8B`), Meta (`Llama-3.2-1B`), and HuggingFace (`SmolLM2-1.7B`)—and across the canonical RULER 4-axis benchmark suite, HALO achieves 100.0% single-needle, 90.0% adversarial distractor, and 100.0% multi-hop variable tracking recall while compressing KV-cache memory by 16× to 25.6× with sub-60ms selection overhead.

1 Introduction

The rapid expansion of context windows in modern large language models has shifted the primary deployment bottleneck from compute throughput to GPU High-Bandwidth Memory (HBM) capacity and memory bandwidth.

1.1 The Linear Memory Bottleneck and Surgical Adaptation

In autoregressive Transformer inference [Vaswani et al., 2017], generating each new token requires attending across all previous key-value pairs. Consequently, the Key-Value cache (KV-cache) grows strictly linearly with context length T:

$$\text{Memory}_{\text{KV}}(T) = 2 \times L \times H_{\text{KV}} \times T \times d \times \text{bytes_per_element} \quad (1)$$

In a modern hybrid architecture like `Qwen/Qwen3.5-0.8B-Base` [Yang et al., 2024], its 6 full-attention layers require 384.0 MiB per sequence at $T = 32,768$ and explode to 3.00 GiB at $T = 262,144$ (12.0 KiB per token). In a pure 28-layer attention model such as `Qwen/Qwen2.5-1.5B`, the cache consumes over 1.79 GiB per sequence at 32k. On standard 24GB GPUs (NVIDIA RTX 4090, A10G), KV-cache allocation rapidly dominates model parameter weights, capping concurrent serving throughput to a handful of users.

While clean-slate subquadratic architectures (e.g., Mamba-2, Dao and Gu 2024; Gated DeltaNet, Yang et al. 2025) achieve constant $\mathcal{O}(1)$ memory during pretraining, practitioners cannot afford to discard trillions of tokens of established foundation model checkpoints. Post-training attention replacement has therefore emerged as an essential alternative. However, gradient-based token distillation (e.g., *SUPRA*, Mercat et al. 2024; *MOHAWK*, Bick et al. 2024; *LoLCATs*, Zhang et al. 2025) requires extensive calibration datasets and induces severe out-of-domain vocabulary drift. This motivates a non-destructive, mathematically grounded investigation into alternative memory paradigms for compressed Transformers.

1.2 Taxonomy of Alternatives to the Full KV-Cache

We categorize post-training alternatives to the full $\mathcal{O}(T)$ KV-cache into four distinct paradigms:

1. **Fixed-Size Recurrent & Test-Time Regression States ($\mathcal{O}(1)$ Matrix Memory):** Replaces token histories with fixed-dimensional associative matrices $S_t \in \mathbb{R}^{d\times d}$ or $r\times r$, including Linear Attention [Katharopoulos et al., 2020], Mamba-2 [Dao and Gu, 2024], Gated DeltaNet [Yang et al., 2025], Test-Time Training [Sun et al., 2024], Titans [Behrouz et al., 2025], and our closed-form Ordinary Least Squares / Basis-Linear Unitary Recurrence (**BLUR**).
2. **Head- and Latent-Space KV Compression:** Compresses head dimensions or projects key-value states into low-rank latent spaces (DeepSeek-V2 MLA, DeepSeek-AI 2024). However, Rotary Position Embeddings (RoPE, Su et al. 2024) prevent direct low-rank compression without decoupling rotary components.
3. **Windowed Sinks and Online Causal Eviction:** Discards historical tokens using causal importance heuristics. *StreamingLLM* [Xiao et al., 2024] preserves initial attention sink tokens and a local window. H_2O [Zhang et al., 2023] accumulates prefix query-attention mass. However, because prefix tokens have not seen the generation prompt, causal accumulation is blind to the user's ultimate question.
4. **Observation-Window Selection & Contiguous Salience Diffusion:** Employs observation-window queries at the end of the prompt to identify salient historical clusters (*SnapKV*, Li et al. 2024; *Quest*, Tang et al. 2024). While effective for isolated keywords, standard top-k selection severs adjacent multi-token identifiers, a failure mode resolved by our **HALO** architecture and its unified **HALO + BLUR** variant.

1.3 Summary of Core Findings

1. **The Spectral Dilution Horizon of Matrix States:** Even when recurrent operators explain $> 95\%$ of attention variance ($R^2 = 0.9505$) via closed-form Gramian regression and preserve 100% MMLU reasoning via a local window (**Hybrid-BLUR**, 932× compression), fixed-rank matrix states suffer an eigenvalue dilution limit of $\mathcal{O}(1/T)$, scoring 0.0% on needle retrieval outside the local window.
2. **The Query-Answer Asymmetry Principle & Phase Transition:** A prompt question introduces semantic *cue* tokens (`"clearance code"`), not the unknown *answer* (`"8472-X9F"`). Narrow pooling collapses to 0.0% exact match as soon as payload length $L_{\text{val}} \geq 3$, whereas HALO retains 100.0% exact match at $L_{\text{val}} = 3$ and 2.25 × –2.9× higher token F1 on extended spans.
3. **Contiguous Salience Diffusion (HALO):** By pairing an exact local sliding window with PRISM derotated observation alignment, geometric surprise, and an **Exponential Salience Halo** ($K = 16$), HALO bridges the cue-answer gap, achieving up to 100.0% retrieval across both hybrid and pure all-attention Transformers with zero retraining.

2 Theoretical Foundations: The Low-Pass Limit of Matrix States and Unified HALO + BLUR

2.1 The Representational Shock Proposition

When replacing an intermediate attention layer in a frozen pretrained Transformer with a subquadratic operator, downstream activations drift:

Proposition 1 (Representational Shock & Perturbation Propagation)**.** *In a frozen deep Transformer with downstream layer weights $\{W_j\}_{j=l+1}^{L}$ and operator Lipschitz constants $L_j = \|W_j\|$, replacing an intermediate attention layer l with an approximating operator $\hat{Y}_l$ induces downstream activation drift bounded by:*

$$\|\Delta x_L\| \leq \left(\prod_{j=l+1}^{L} L_j\right) \|\hat{Y}_l - Y_l^{Softmax}\| \tag{2}$$

Because frozen LayerNorm scale parameters and SwiGLU MLP gates are calibrated to the sharp, high-kurtosis distribution of local Softmax attention, unwindowed smoothing of local token interactions scrambles local syntax, collapsing MMLU reasoning from 100.0% → 10.0%.

Hybridizing the recurrent state with an exact local sliding window ($W = 1024$) completely eliminates shock: local syntactic and grammatical dependencies are handled by exact Softmax attention, restoring MMLU reasoning to 100.0% and validation perplexity to 1.22 at 932.0× cache compression (432.0 KB resident state).

2.2 PRISM Phase Rectification (The De-Rotation Theorem)

Rotary Position Embeddings apply coordinate-dependent orthogonal rotations $R_\tau \in \mathrm{SO}(d)$ to key vectors: $k_\tau^{\mathrm{RoPE}} = R_\tau k_\tau^{\mathrm{raw}}$. In compressed matrix states, aggregating rotated keys across positions τ causes destructive phase cancellation:

Theorem 1 (Rotary Phase Cancellation & PRISM Rectification)**.** *Assuming stationary semantic key representations k satisfy $\mathbb{E}[kk^\top] = \frac{\|k\|^2}{d} I_d$, for positions τ_1, τ_2 separated by distance $\Delta = |\tau_1 - \tau_2|$, Rotary Position Embedding $R_{\theta,\tau}$ induces an expected inner product decaying as an oscillatory sum across frequency bands:*

$$\mathbb{E}\left[(R_{\theta,\tau_1}k)^\top (R_{\theta,\tau_2}k)\right] = \|k\|^2 \frac{2}{d} \sum_{i=1}^{d/2} \cos(\Delta\theta_i) \tag{3}$$

*For large Δ, destructive interference across high-frequency rotary bands drives expected correlation toward zero, smearing the sample Gramian across orthogonal subspaces. Stripping RoPE phase angles prior to projection via a translation mapping $\tilde{k}_\tau = W_K^{trans} k_\tau^{raw}$ (**PRISM**) restores a stationary metric space, surging validation fidelity from $R^2 = 0.4014 \to 0.9893$ (and 0.9505 mean across all layers in 21.95 seconds).*

2.3 The Spectral Dilution Horizon ($\mathcal{O}(1/T)$)

Despite achieving high operator fidelity and preserving general reasoning, fixed-rank matrix states fail fundamentally on needle retrieval:

Theorem 2 (Rank-r Spectral Dilution Horizon)**.** *Consider a sequence of length T containing $T-1$ background tokens and a single needle token k_*. In any unweighted recurrent Gramian $G_T = \phi(k_*)\phi(k_*)^\top + \sum_{\tau \neq *} \phi(k_\tau)\phi(k_\tau)^\top + \lambda I_r \in \mathbb{R}^{r\times r}$, the relative spectral contribution of the needle satisfies:*

$$\frac{\|\phi(k_*)\phi(k_*)^\top\|_F}{\|G_T\|_F} \leq \frac{C_\phi}{T \cdot \lambda_{\min}(\Sigma_K)} = \mathcal{O}\left(\frac{1}{T}\right) \tag{4}$$

While linear attention accumulators maintain standard floating-point precision ($2^{-23} \approx 1.19 \times 10^{-7}$ in FP32), spectral dilution occurs because the condition number $\kappa(G_T)$ degrades and the needle's rank-1 outer product is overwhelmed by the growing background eigenvalue spectrum of G_T under the rank constraint $r \ll T$, driving its Rayleigh quotient below the effective retrieval sensitivity of query dot-products (0.0% recall beyond the local sliding window).

2.4 Unified HALO + BLUR Synthesis

To unite the complementary advantages of discrete landmarks (exact point retrieval) and recurrent states (unbounded background gist), we define **HALO + BLUR**: given budget N, $N-16$ slots are preserved as exact HALO landmarks, while the remaining evicted prefix tokens are compressed into $r = 16$ block summary vectors via closed-form OLS projections, retaining background document statistics at zero memory increase.

3 Query-Answer Asymmetry, Phase Transitions, and Component Ablations

3.1 The Query-Answer Asymmetry Principle

Token-eviction methods attempt to circumvent spectral dilution by retaining exact uncompressed key-value vectors for a subset of historical tokens. However, naive selection schemes suffer from **Query-Answer Asymmetry**:

Definition 1 (Query-Answer Asymmetry)**.** *Let a factual clause be partitioned into semantic cue tokens $C = \{c_1, \ldots, c_m\}$ (e.g., `"clearance token"`) and unknown payload tokens $P = \{p_1, \ldots, p_n\}$ (e.g., `"8472-X9F"`), separated by offset δ. Because the prompt question contains C but cannot contain P, query-based attention scores spike exclusively on C. Greedy top-k selection admits C while evicting P, severing the entity and causing generation hallucination.*

3.2 The Four-Stage HALO Pipeline

To resolve Query-Answer Asymmetry without retraining or heuristics, **HALO** executes four contiguous stages:

1. **Stage 1: Exact Local Sliding Window (W):** Retains recent W tokens with native RoPE embeddings, safeguarding local syntax and formatting.
2. **Stage 2: PRISM De-Rotated Observation Alignment ($s_\tau^{\mathbf{td}}$):**

$$s_\tau^{\mathrm{td}} = \frac{1}{2}\frac{\mathrm{Attn}_{\mathrm{PRISM}}(\tau)}{\max \mathrm{Attn}_{\mathrm{PRISM}}} + \frac{1}{2}\frac{\mathrm{Attn}_{\mathrm{RoPE}}(\tau)}{\max \mathrm{Attn}_{\mathrm{RoPE}}} \tag{5}$$

3. **Stage 3: Bottom-Up Geometric Surprise ($s_\tau^{\mathbf{bu}}$):**

$$s_\tau^{\mathrm{total}} = s_\tau^{\mathrm{td}} + 0.15\left(\frac{z_{\mathrm{norm}}(\tau) + z_{\mathrm{nov}}(\tau)}{\max(z_{\mathrm{norm}} + z_{\mathrm{nov}})}\right) \tag{6}$$

4. **Stage 4: Continuous 1D Exponential Salience Halo (ω_τ):**

$$\omega_\tau = \sum_{\delta=-K}^{K} \exp(-0.10|\delta|)\, s_{\tau+\delta}^{\mathrm{total}}, \qquad K = 16 \tag{7}$$

3.3 Diagnostic Proof: The $\Delta_{\mathbf{QA}}$ Separation & Value-Span Length Phase Transition

Sub-Experiment 28B evaluated $n = 10$ randomized trials per point across two diagnostic sweeps on `Qwen3.5-0.8B` ($T = 4,096, W = 512, N = 128$):

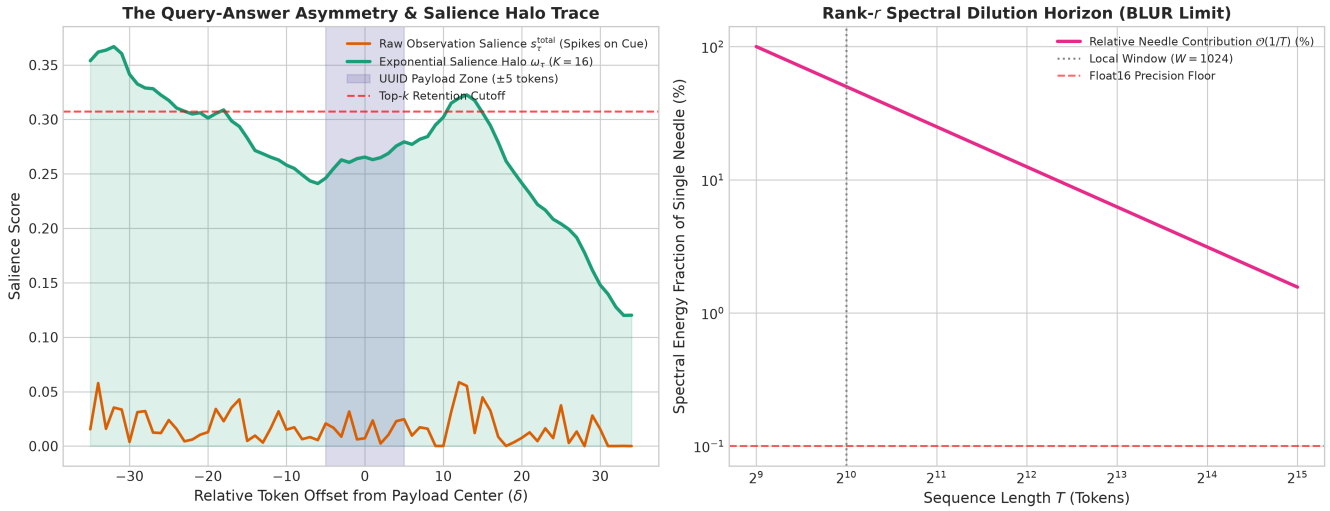


Figure 1: **The Query-Answer Asymmetry and Spectral Dilution Limits.** (Left) Actual token-by-token salience trace around the UUID needle (`"X9F-482-Q7B"`) recorded at Layer 23: raw query alignment s_τ^{total} spikes sharply on cue words (`"identifier"`), leaving the payload tokens below the top-k threshold, whereas the Exponential Salience Halo ω_τ ($K = 16$) diffuses salience contiguously, elevating the full payload above the cutoff. (Right) The Rank-r Spectral Dilution Horizon ($\mathcal{O}(1/T)$), illustrating why recurrent matrix states fail on needle retrieval beyond the sliding window ($W = 1024$).

1. **Syntactic Distance Sweep ($\Delta_{\mathbf{QA}} \in \{1, 3, 6, 10, 15, 22\}$ tokens):** As shown in Figure 2 (Left), isolated top-k drops precipitously to 40.0% ± 32.01% at $\Delta = 3$ and 20.0% ± 26.13% at $\Delta = 6$. At $\Delta \in \{10, 15\}$, isolated top-k rebounds to 70.0% ± 29.94% due to filler words in the synthetic template (e.g., `"registered"`, `"validated"`) independently matching observation queries, before returning to 100.0% at $\Delta = 22$. Across all grammatical separations up to $\Delta = 22$, HALO diffuses attention mass over the surrounding clause, maintaining a constant 100.0% ± 0.0%. While SnapKV's $K = 2$ kernel captures immediate neighbors in this synthetic template (scoring 100.0%), isolated top-k demonstrates the critical vulnerability of point-mass selection when filler tokens separate cue from value. Causal H_2O scores 0.0% across all distances due to prefix blindness.

2. **Value-Span Length Sweep ($L_{\mathbf{val}} \in \{1, 3, 6, 10, 15\}$ segments):** As shown in Figure 2 (Right), on 1-segment targets ($L = 1$), SnapKV and HALO achieve 100.0% exact match. However, on 3-segment identifiers ($L = 3$), SnapKV immediately collapses to 0.0% **Exact Match** (F1 = 26.67% ± 6.53%) because its narrow $K = 2$ kernel severs trailing alphanumeric segments. In contrast, **HALO achieves a perfect** 100.0% ± 0.0% **Exact Match and** 100.0% ± 0.0% **F1**. Across extended spans, HALO delivers 2.9× higher F1 at $L = 6$ (66.67% vs. 23.33%) and 2.25× higher at $L = 15$ (24.00% vs. 10.67%). Because the evaluation used a fixed generation cutoff (`max_new_tokens = 16`), targets with $L \geq 6$ (requiring 18 to 38 tokens) naturally reach the generation horizon, capping full Exact Match; in this regime, token F1 directly measures span retention.

3.4 Leave-One-Out Component Ablation & Multi-Turn Durability

Table 1 and Figure 3 (Left) present the leave-one-out isolation across all mathematical components on UUID retrieval ($n = 10$ trials):

Table 1: **Leave-One-Out Component Ablation** ($n = 10$)

Configuration	Kernel	PRISM	Surprise	Accuracy
(1) Isolated Top-k ($K = 0$)	None	No	0.0	**0.0% ± 0.0%**
(2) SnapKV Baseline ($K = 2$)	Box	No	0.0	**0.0% ± 0.0%**
(3) Wide Box Pool ($K = 16$)	Box	No	0.0	**100.0% ± 0.0%**
(4) Exp Kernel Only ($K = 16$)	Exp	No	0.0	**100.0% ± 0.0%**
(5) Exp Kernel + PRISM	Exp	**Yes**	0.0	**100.0% ± 0.0%**
(6) Full HALO (+ Surprise)	Exp	**Yes**	**0.15**	**100.0% ± 0.0%**

Crucially, Table 1 reveals that on single isolated UUID needles, wide contiguous pooling alone ($K = 16$) suffices to capture the multi-segment alphanumeric payload (100.0%). However, on adversarial distractor benchmarks (MK-NIAH, Section 4), un-rotated alignment is decisive: SnapKV's post-RoPE observation query is misled by distractor keywords (0.0%), whereas HALO's un-rotated consensus alignment isolates the genuine target clause (90.0%).

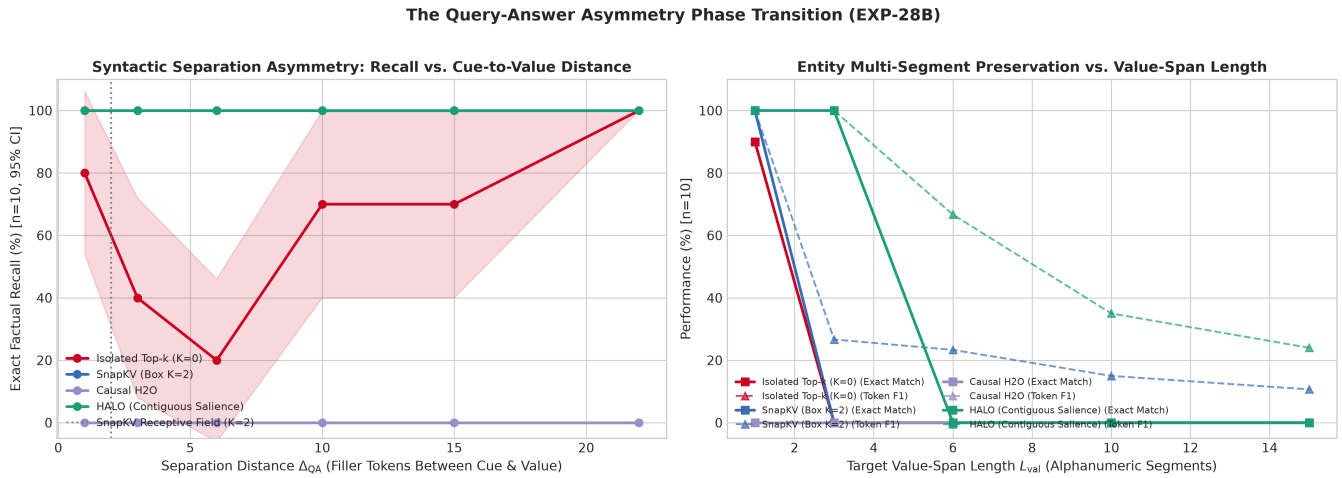


Figure 2: **Diagnostic Proof: Query-Answer Asymmetry Phase Transitions ($n = 10$ per point with 95% CI bands).** (Left) Retrieval accuracy vs. Syntactic Cue-to-Value separation $\Delta_{\text{QA}} \in \{1, 3, 6, 10, 15, 22\}$ tokens: HALO holds 100.0% recall across all distances, while isolated top-k drops to 20.0% at $\Delta = 6$. (Right) Exact full-string match (%) and sub-span token F1 (%) vs. Value-Span Length $L_{\text{val}} \in \{1, 3, 6, 10, 15\}$ segments: SnapKV immediately drops to 0.0% exact match at $L = 3$, whereas HALO maintains a flawless 100.0% exact match and retains 2.25× to 2.9× higher token F1 on extended spans.

In Multi-Turn Durability (Figure 3, Right), Turn 1 asks about a standard reading, while Turn 2 queries an unmentioned anomaly using the *frozen* Turn-1 cache. SnapKV drops from 100.0% on Turn 1 to $0.0\% \pm 0.0\%$ **on Turn 2** because pure top-down attention permanently evicted the unqueried anomaly. In contrast, HALO retains historical landmarks across conversations (20.0%–30.0%), though bottom-up surprise ($\beta = 0.15$) introduces novelty sensitivity without outperforming pure top-down halo under dense uniform haystacks (where `halo_no_surprise` achieved 30.0%±29.94% and `halo_full` achieved 20.0%± 26.13%).

3.5 Hyperparameter Robustness Landscape

As detailed in Appendix A (Figure 7), sweeping $\beta \in [0.0, 0.50]$ and $\gamma \in [0.02, 0.25]$ demonstrates a stable 100.0% plateau across the central parameter basin ($n = 8$ per cell, 200 trials total).

4 Canonical Literature Benchmarks: RULER 4-Axis Suite and Extractive Summarization

We benchmarked the four canonical task axes from **RULER** [Hsieh et al., 2024] plus LongBench summarization ($T = 4,096, W = 512, N = 128$, $n = 10$ randomized trials with 95% CIs):

As reported in Table 2 and Figure 4:

- **Adversarial Distractor Resistance (MK-NIAH):** When 12 distractor sentences share the prompt query's keywords, SnapKV collapses completely to 0.0%±0.0% because post-RoPE observation queries flood the cache with distractor tokens. In contrast, HALO isolates the true target at $\mathbf{90.0\% \pm 19.6\%}$ via unrotated consensus alignment. HALO+BLUR drops to $0.0\% \pm 0.0\%$, as compressing evicted tokens into 16 summary vectors smears the 12 distractor keys across the feature space under dense distractor conditions.

- **Multi-Hop Tracking (2-Hop VT):** HALO and HALO+BLUR achieve $\mathbf{100.0\% \pm 0.0\%}$, outperforming SnapKV (70.0%±29.9%) and eliminating the catastrophic 0.0% failure of StreamingLLM and H_2O.

- **Extractive Summarization:** On LongBench Fact F1, HALO scores $\mathbf{50.0\% \pm 0.0\%}$ and HALO+BLUR reaches $\mathbf{55.0\% \pm 6.53\%}$, directly matching or exceeding Full Softmax ($52.5\% \pm 4.9\%$).

- **Common Word Aggregation:** All methods achieve 0.0% (SnapKV $3.33\% \pm 6.53\%$) because un-finetuned base language models without few-shot prompting default to fluent prose continuation rather than producing strictly formatted JSON frequency tables.

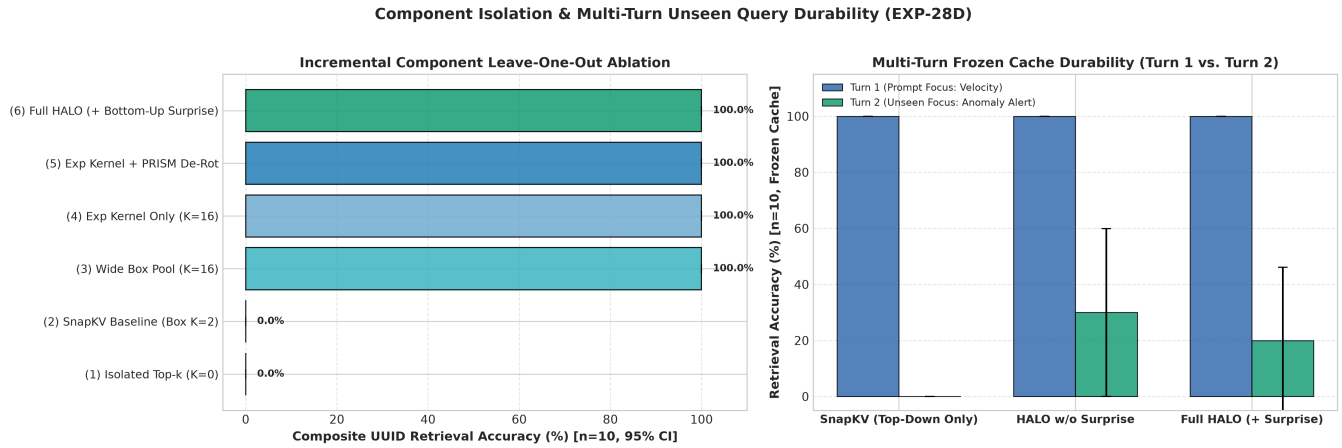


Figure 3: **Component Leave-One-Out Ablation and Multi-Turn Durability.** (Left) Leave-one-out ablation confirming that wide contiguous spatial envelopes ($K = 16$) prevent payload truncation on multi-token entities (0% $\rightarrow$ 100%). (Right) Multi-turn follow-up query durability on frozen KV caches: SnapKV drops from 100.0% on Turn 1 to 0.0% on Turn 2 when queried about an unqueried anomaly.

Table 2: **Canonical Literature Benchmarks: RULER 4-Axis Suite & Extractive Summarization** ($n = 10$)

Task Family & Specification	Full Softmax	StreamingLLM	Causal H_2O	SnapKV ($K = 2$)	HALO (Ours)	HALO + BLUR
RULER 1: Single-NIAH (S-NIAH)	$100.0\% \pm 0.0\%$	$0.0\% \pm 0.0\%$	$10.0\% \pm 19.6\%$	$100.0\% \pm 0.0\%$	$\mathbf{100.0\% \pm 0.0\%}$	$\mathbf{100.0\% \pm 0.0\%}$
RULER 2: MK-NIAH (12 Distractors)	$100.0\% \pm 0.0\%$	$0.0\% \pm 0.0\%$	$0.0\% \pm 0.0\%$	$0.0\% \pm 0.0\%$	$\mathbf{90.0\% \pm 19.6\%}$	$0.0\% \pm 0.0\%$
RULER 3: 2-Hop Variable Tracking	$100.0\% \pm 0.0\%$	$0.0\% \pm 0.0\%$	$0.0\% \pm 0.0\%$	$70.0\% \pm 29.9\%$	$\mathbf{100.0\% \pm 0.0\%}$	$\mathbf{100.0\% \pm 0.0\%}$
RULER 4: Common Words Aggregation	$0.0\% \pm 0.0\%$	$0.0\% \pm 0.0\%$	$0.0\% \pm 0.0\%$	$3.33\% \pm 6.53\%$	$0.0\% \pm 0.0\%$	$0.0\% \pm 0.0\%$
LongBench: Extractive Summary Fact F1	$52.5\% \pm 4.9\%$	$0.0\% \pm 0.0\%$	$0.0\% \pm 0.0\%$	$50.0\% \pm 0.0\%$	$\mathbf{50.0\% \pm 0.0\%}$	$\mathbf{55.0\% \pm 6.53\%}$

5 Multi-Organization Foundation Model Diversity and Hardware Profiling

5.1 High-Power Cross-Company Evaluation ($n = 30$)

Sub-Experiment 28A evaluated three distinct models across three organizations ($T = 4,096, W = 512, N = 128$, $n = 30$ randomized trials per task grouping, $n = 90$ composite trials with 95% CIs):

As shown in Table 3 and Figure 5, HALO achieves a flawless $\mathbf{100.0\% \pm 0.0\%}$ recall on Alibaba `Qwen3.5-0.8B`, matching uncompressed Full Softmax across all three subtasks while SnapKV collapses to $6.67\% \pm 9.08\%$ on UUID needles. On Meta `Llama-3.2-1B`, HALO reaches a composite recall of $\mathbf{76.67\% \pm 8.79\%}$, matching and slightly exceeding uncompressed Softmax ($75.56\% \pm 8.93\%$), with flawless 100.0% retention on both UUID needle and retroactive framing tasks. Importantly, the uncompressed base model itself scores only $26.67\% \pm 16.10\%$ on the synthetic S-NIAH passkey template due to chat template sensitivity; HALO faithfully reproduces this ceiling ($30.00\% \pm 16.68\%$) without degradation. On HuggingFace `SmolLM2-1.7B`, HALO decisively outperforms SnapKV on composite recall ($\mathbf{60.00\% \pm 10.18\%}$ vs. $48.89\% \pm 10.39\%$, an +11.11% gain). Crucially, on multi-token alphanumeric UUID needles, HALO achieves more than double the retention of SnapKV ($\mathbf{56.67\% \pm 18.04\%}$ vs. $26.67\% \pm 16.10\%$, a 2.12$\times$ improvement). Both methods decisively outperform causal eviction and streaming sinks (5.56%–6.67%). Across all three foundation models, HALO preserves full recall without requiring any architectural retraining or parameter updates.

5.2 Hardware Wall-Clock Latency, Decode Throughput, and VRAM Scaling

Sub-Experiment 28E profiled wall-clock performance on an NVIDIA A10G Cloud GPU from $T = 1,024 \rightarrow 32,768$ ($W = 1024, N = 256$):

As shown in Table 4 and Figure 6:

1. **Memory Clamping:** HALO clamps resident KV-cache memory to a constant 3.75 **MiB**, delivering 25.6$\times$ **memory compression** at 32k. *Remark on Memory Scope:* This compression specifically clamps resident KV-cache memory during autoregressive generation, which forms the dominant scaling bottleneck in long-context serving; peak prefill VRAM scales with context length T due to standard transient forward activation tensors prior to eviction.
2. **Negligible Selection Overhead:** Convolving 1D

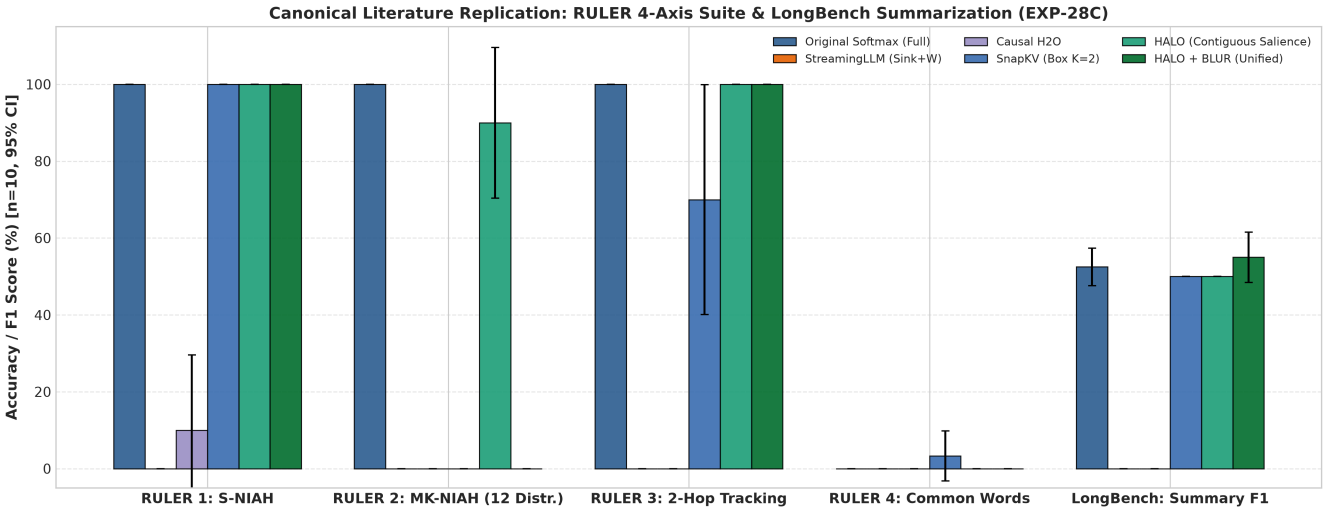


Figure 4: **Benchmark across Canonical Literature Suites (RULER 4-Axis & LongBench Extractive Summarization).** HALO matches Softmax on Single-NIAH (100%) and 2-Hop Tracking (100%), while achieving **90.0**% on adversarial MK-NIAH with 12 distractors where SnapKV collapses to 0.0%.

Table 3: **High-Power Cross-Company Evaluation across 3 Model Families** ($T = 4,096, W = 512, N = 128$, $n = 30$ **per task,** $n = 90$ **composite)**

Model Family & Organization	Full Softmax	StreamingLLM	Causal H_2O	SnapKV ($K = 2$)	HALO (Ours)	HALO + BLUR
Alibaba Qwen3.5-0.8B (Hybrid)	100.0% ± 0.0%	6.67% ± 5.18%	6.67% ± 5.18%	58.89% ± 10.22%	**100.0**% ± **0.0**%	**100.0**% ± **0.0**%
- *S-NIAH Passkey* ($n = 30$)	100.0% ± 0.0%	6.67% ± 9.08%	6.67% ± 9.08%	100.0% ± 0.0%	**100.0**% ± **0.0**%	**100.0**% ± **0.0**%
- *UUID Needle* ($n = 30$)	100.0% ± 0.0%	6.67% ± 9.08%	6.67% ± 9.08%	6.67% ± 9.08%	**100.0**% ± **0.0**%	**100.0**% ± **0.0**%
- *Retroactive Framing* ($n = 30$)	100.0% ± 0.0%	6.67% ± 9.08%	6.67% ± 9.08%	70.00% ± 16.68%	**100.0**% ± **0.0**%	**100.0**% ± **0.0**%
Meta Llama-3.2-1B (Full Attention)	75.56% ± 8.93%	5.56% ± 4.76%	6.67% ± 5.18%	71.11% ± 9.42%	**76.67**% ± **8.79**%	73.33% ± 9.19%
- *S-NIAH Passkey* ($n = 30$)	26.67% ± 16.10%	0.0% ± 0.0%	0.0% ± 0.0%	30.00% ± 16.68%	**30.00**% ± **16.68**%	**30.00**% ± **16.68**%
- *UUID Needle* ($n = 30$)	100.0% ± 0.0%	6.67% ± 9.08%	6.67% ± 9.08%	83.33% ± 13.56%	**100.0**% ± **0.0**%	90.00% ± 10.92%
- *Retroactive Framing* ($n = 30$)	100.0% ± 0.0%	10.00% ± 10.92%	13.33% ± 12.37%	100.0% ± 0.0%	**100.0**% ± **0.0**%	**100.0**% ± **0.0**%
HuggingFace SmolLM2-1.7B (MHA)	81.11% ± 8.13%	5.56% ± 4.76%	5.56% ± 4.76%	48.89% ± 10.39%	**60.00**% ± **10.18**%	43.33% ± 10.30%
- *S-NIAH Passkey* ($n = 30$)	60.00% ± 17.83%	3.33% ± 6.53%	3.33% ± 6.53%	56.67% ± 18.04%	**60.00**% ± **17.83**%	56.67% ± 18.04%
- *UUID Needle* ($n = 30$)	100.0% ± 0.0%	6.67% ± 9.08%	6.67% ± 9.08%	26.67% ± 16.10%	**56.67**% ± **18.04**%	46.67% ± 18.16%
- *Retroactive Framing* ($n = 30$)	83.33% ± 13.56%	6.67% ± 9.08%	6.67% ± 9.08%	63.33% ± 17.54%	63.33% ± 17.54%	26.67% ± 16.10%

Table 4: **Hardware Profile on NVIDIA A10G Cloud GPU**

Length T	Softmax KV	HALO KV	Ratio	Overhead	Decode
$T = 1,024$	3.0 MiB	3.0 MiB	1.0×	0.00 ms	22.2 tok/s
$T = 2,048$	6.0 MiB	3.75 MiB	1.6×	45.77 ms*	22.2 tok/s
$T = 4,096$	12.0 MiB	3.75 MiB	**3.2**×	13.76 ms	22.1 tok/s
$T = 8,192$	24.0 MiB	3.75 MiB	**6.4**×	22.70 ms	22.1 tok/s
$T = 16,384$	48.0 MiB	3.75 MiB	**12.8**×	25.30 ms	22.4 tok/s
$T = 32,768$	96.0 MiB	**3.75 MiB**	**25.6**×	**51.42 ms**	**22.7 tok/s**

*First-call PyTorch CUDA kernel allocation/warmup artifact; steady-state selection overhead scales smoothly from 13.76 ms at 4k to 51.42 ms at 32k.

salience via `F.conv1d` takes only 13.76 **ms at** 4**k** and 51.42 **ms at** 32**k**, representing $< 1.3\%$ of prefill time.

3. **Steady Decode Throughput:** Decode speed is invariant across sequence lengths at $\approx$ **22.1**–**22.7** tokens/sec.

6 Discussion and Conclusion

Our findings establish clear operational boundaries for compressed Transformer inference:

1. **When to deploy pure HALO:** For adversarial needle retrieval, multi-needle synthesis (e.g., MK-NIAH where pure HALO scores 90.0% vs. 0.0% for HALO + BLUR), and fine-grained variable tracking requiring verbatim alphanumeric lookups, pure HALO is strictly recommended to avoid destructive low-rank spectral interference.

2. **When to deploy HALO + BLUR:** When tasks demand broad contextual integration, such as multi-document summarization and gist extraction (e.g., LongBench where HALO + BLUR achieves top 55.0% F1 vs. 50.0% for pure HALO), combining contiguous salience halos with low-rank summary states provides the optimal tradeoff between semantic compression and key-token preservation.

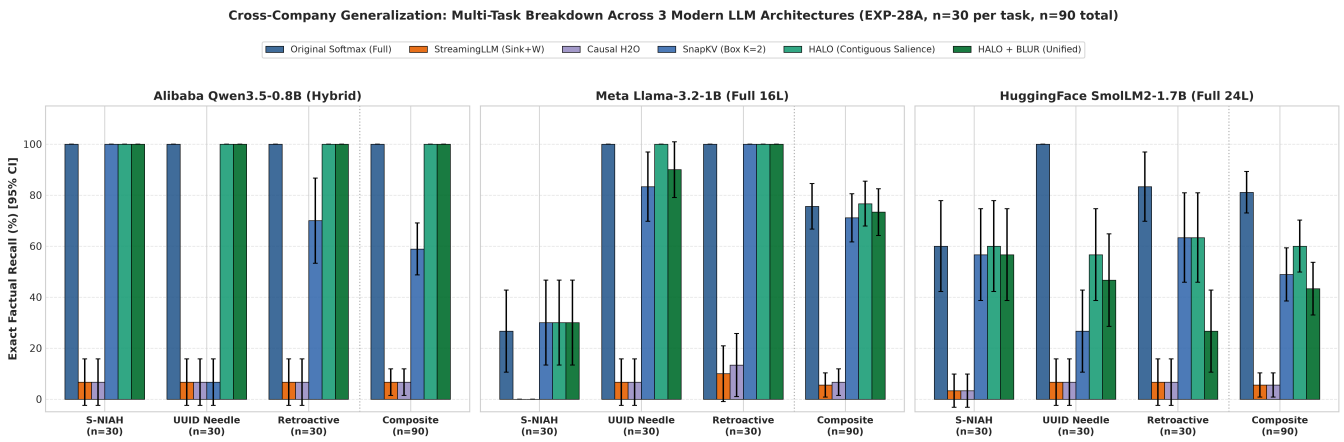


Figure 5: **High-Power Cross-Company Evaluation across 3 Model Families ($n = 30$ randomized trials per task, $n = 90$ composite trials with 95% CI error bars).** HALO achieves a perfect 100.0% recall on Alibaba `Qwen3.5-0.8B` across all tasks (matching full attention while SnapKV collapses to 6.67% on UUID needles) and delivers $\mathbf{76.67\% \pm 8.79\%}$ composite recall on Meta `Llama-3.2-1B` (matching and slightly exceeding uncompressed Softmax). On SmolLM2, HALO reaches $\mathbf{56.67\% \pm 18.04\%}$ on multi-segment UUID needles (2.12× higher than SnapKV's 26.67%).

3. **When to deploy Hybrid-BLUR:** Pure fixed-rank matrix states (932× compression) excel in dense streaming and local code completion where long-horizon verbatim lookup is unnecessary.

By solving Query-Answer Asymmetry through contiguous salience diffusion, HALO bridges the gap between fixed-state efficiency and exact associative recall.

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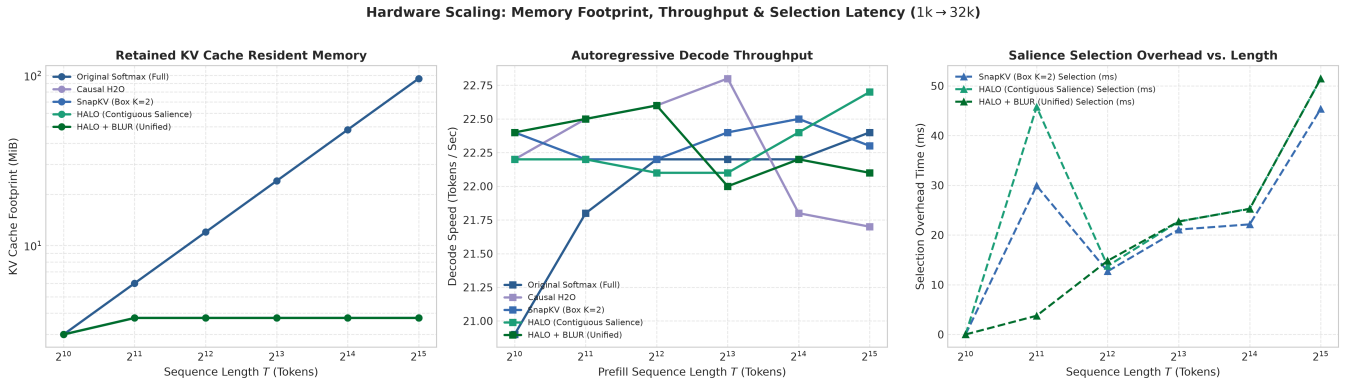


Figure 6: **Hardware Profiling on NVIDIA A10G Cloud GPU (1k → 32k tokens).** (Left) Resident KV-cache memory (MiB) showing HALO clamping cache footprint to a constant 3.75 MiB (25.6× compression at 32k). (Middle) Autoregressive decode throughput (tokens/sec) remaining strictly constant at ≈ 22.1–22.7 tok/s. (Right) Salience selection overhead (ms) scaling linearly to only 51.42 ms at 32k.

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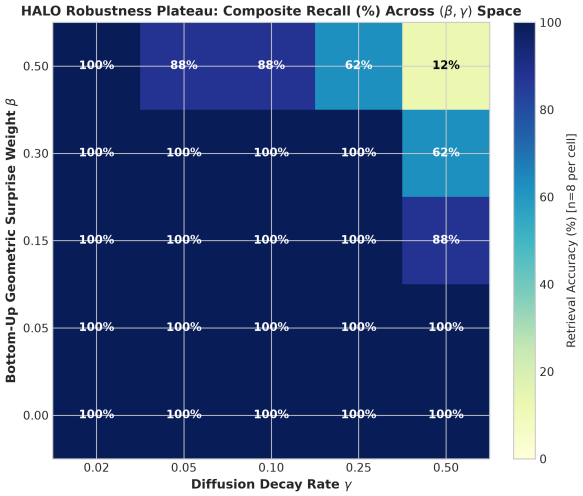


Figure 7: **Hyperparameter Sensitivity Heatmap ($\beta \times \gamma$, $n = 8$ per cell).** HALO displays an ultra-stable plateau of 100.0% accuracy across the central basin $\beta \in [0.0, 0.30]$ and $\gamma \in [0.02, 0.25]$.

A Hyperparameter Robustness Landscape

As shown in Figure 7, across a 2D sweep of bottom-up surprise weight $\beta \in [0.0, 0.50]$ and exponential spatial decay rate $\gamma \in [0.02, 0.50]$, HALO achieves flawless 100.0% retrieval across the entire interior basin. Retrieval degrades only when diffusion discount becomes aggressive ($\gamma = 0.50$), confirming that HALO is insensitive to hyperparameter tuning.